INCH-POUND

MIL-PRF-1/1261G
22 July 1999
SUPERSEDING
MIL-E-1/1261F
21 March 1972

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY
TYPES 12ABP1A, 12ABP7A, AND 12ABP19A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Magnetic deflection, electrostatic focus, aluminized screen.

PIN CONNECTIONS AND DIMENSIONS. See figure 1.

ABSOLUTE RATINGS:

| Parameter: | Ef | Ec1 | Ec2 | Eb2 | Ehk | Rg1 | Eb1 | Alt |
|---|---|---|---|---|---|---|---|---|
| Unit: | V | V dc | V dc | V dc | V dc | Meg | V dc | ft |
| Maximum: | 6.9 | 0, -200 | 770 | 13,200 | ±180 | 1/ | 1,100 | 10,000 |
| Minimum: | 5.7 | --- | --- | 7,000 | --- | --- | -500 | --- |
| Test conditions: | 6.3 | Adjust | 300 | 10,000 | --- | --- | Focus | --- |

See footnotes at end of table I.

GENERAL:

Qualification: Required.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:20Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection.

| Inspection | Method | Type | Conditions | Acceptance level | Inspection level or code | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Min | Max | |
| Qualification | | | | | | | | | |
| Base material insulating quality | 1216 | All | Zone 5 (minimum) | --- | --- | --- | --- | --- | --- |
| Pressure | 1141 | All | | --- | --- | --- | --- | --- | --- |
| Cathode illumination | 5216 | All | | --- | --- | --- | --- | --- | --- |
| Vibration (cathode-ray tubes) | 5111 | All | | --- | --- | Width | --- | 2.0 | mm |
| Direct interelectrode capacitance | 1331 | All | Ck to all<br>Cg1 to all | ---<br>--- | ---<br>--- | C<br>C | ---<br>--- | 6.0<br>10.0 | pF<br>pF |
| Face tilt | 5101 | All | | --- | --- | --- | --- | --- | --- |
| Shock | --- | All | 2/ | --- | --- | --- | --- | --- | --- |
| Conformance inspection, part 1 | | | | | | | | | |
| Bulb, screen, and faceplate quality | 5106 | All | | --- | --- | --- | --- | --- | --- |
| Voltage breakdown | 5201 | All | | --- | --- | --- | --- | --- | --- |
| Voltage breakdown (magnetic type) | 5201 | All | | --- | --- | --- | --- | --- | --- |
| Gas ratio | 5206 | All | 7/ | --- | --- | Gr | --- | 0.25 | --- |
| Neck straightness | 5101 | All | 3/ | --- | --- | --- | --- | --- | --- |
| Light output | 5221 | P1A | Ib = 100 µA dc | --- | --- | --- | 125 | --- | fL |
| Modulation | 5223 | P1A, P7A | Ib = 100 µA dc | --- | --- | ΔEc1 | --- | 32 | V dc |
| | | P19A | Ib2 = 2 µA dc | --- | --- | ΔEc1 | --- | 15 | V dc |
| Spot position (magnetic deflection) | 5231 | All | | --- | --- | --- | --- | 15 | mm |
| Zero-bias anode current (magnetic deflection) | 5236 | P1A, P7A | | --- | --- | --- | --- | --- | --- |
| Grid cutoff voltage | 5241 | All | | --- | --- | Ec1 | -33 | -77 | V dc |
| Grid No. 1 leakage | 5251 | All | | --- | --- | --- | --- | --- | --- |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:20Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

| Inspection | Method | Type | Conditions | Acceptance level | Inspection level or code | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Min | Max | |
| Conformance inspection, part 2 | | | | | | | | | |
| Permanence of marking | 1105 | All | | --- | --- | --- | --- | --- | --- |
| Secureness of base, base insert, and cap | 1101 | All | | --- | --- | --- | --- | --- | --- |
| Heater current | 1301 | All | | --- | --- | If | 540 | 660 | mA |
| Side terminal and base alignment | 5101 | All | Vacant pin position No. 3 | --- | --- | --- | --- | --- | --- |
| Base pin solder depth | 1111 | All | | --- | --- | --- | --- | --- | --- |
| Neck and bulb alignment (magnetic type) | 5101 | All | | --- | --- | --- | --- | --- | --- |
| Focusing voltage at beam current condition | --- | P1A, P7A | Ib2 = 100 µA dc 4/ | --- | --- | --- | 0 | 300 | V dc |
| | | P19A | Ib2 = 2 µA dc 5/ | --- | --- | --- | 0 | 400 | V dc |
| Screens | 5221 | P7A, P19A | | --- | --- | --- | --- | --- | --- |
| Stray emission (conventional types) | 5216 | All | Eb2 = 13,200 V dc; Ec2 = 770 V dc; Eb1 = 1,100 V dc | --- | --- | --- | --- | --- | --- |
| Line width A (magnetic deflection) | 5226 | P1A, P7A | Ib2 = 100 µA dc 6/ | --- | --- | --- | --- | 0.50 | mm |
| | | P19A | Ib2 = 2 µA dc 6/ | --- | --- | --- | --- | 0.35 | mm |
| Heater-cathode leakage | 5251 | All | | --- | --- | --- | --- | 15 | µA |
| Electrode currents (grid No. 2) | 5251 | All | Ec1 = 0 | --- | --- | Ic2 | -15 | +15 | µA dc |
| Grid No. 2 leakage | 5251 | All | | --- | --- | --- | --- | --- | --- |
| Conformance inspection, part 3 | | | | | | | | | |
| Life-test provisions | --- | All | Group C; t = 500 hours; Eb2 = 13,200 V dc; Ec2 = 770 V dc; Eb1 focus | --- | --- | --- | --- | --- | --- |
| | | P1A, P7A | Ib2 = 60 µA dc | --- | --- | --- | --- | --- | --- |
| | | P19A | Ib2 = 2 µA dc | --- | --- | --- | --- | --- | --- |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:20Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

| Inspection | Method | Type | Conditions | Acceptance level | Inspection level or code | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Min | Max | |
| Conformance inspection, part 3 - Continued | | | | | | | | | |
| Cathode ray tubes, life-test end points: | --- | | | | | | | | |
| Line width A | | P1A, P7A | | --- | --- | --- | --- | 0.50 | mm |
| Line width A | | P19A | | --- | --- | --- | --- | 0.35 | mm |
| Modulation | | P1A, P7A | | --- | --- | --- | --- | 32 | V dc |
| Modulation | | P19A | | --- | --- | --- | --- | 15 | V dc |
| Grid No. 1 leakage | | All | | --- | --- | --- | --- | ---- | --- |
| Heater-cathode leakage | | | | --- | --- | --- | --- | ---- | --- |
| Grid No. 2 leakage | | | | --- | --- | --- | --- | ---- | --- |
| Stray emission | | | | --- | --- | --- | --- | ---- | --- |

1/ When Ec2 is greater than 330 V dc, Rgl shall not exceed 0.5 meg. When Ec2 is less than 330 V dc, Rg1 shall not exceed 1.5 meg.

2/ The test shall be performed in accordance with method 205, test condition A, of MIL-STD-202.

(a) The following conditions apply:

(1) The tube shall be mounted to the elevator table by a suitable rigid assembly. Hardwood clamps shall be used at the face plate and near the neck to secure the tubes to the assembly.

(2) The assembly shall be such as to permit changes in tube orientation. A total of 10 shocks shall be applied to the tube in the following sequences:

| Order | Number of drops | Direction |
|---|---|---|
| 1 | 2 | Base to face |
| 2 | 2 | x+ |
| 3 | 2 | x- |
| 4 | 2 | y+ |
| 5 | 2 | y- |

(3) No potentials shall be applied during shock.

(b) After shock test, cathode-ray tubes shall meet the requirements specified and shall be subjected to the following tests:

(1) The change in position of the undeflected focused spot from its position noted before the test (see method 5231) shall not exceed 8 mm.

(2) The change in deflection factor for electrostatic tube types shall not exceed 5 percent (see method 5248).

(3) Starting with a tube with no loose particles, the maximum number of loose particles as a result of shocking shall be no more than a total of five loose particles greater than .016 inch (0.41 mm) in any one cathode-ray tube. Tubes having loose particles greater than .016 inch (0.41 mm) shall be vibrated with the face down in a horizontal plate at 5 G's for 1 minute. If the screen shows damage from such vibration, the tube shall be rejected.

(4) The change in grid cutoff voltage shall not exceed 10 percent (see method 5241).

(5) There shall be no damage to the external parts of the tube.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:20Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

3/ The cylinder shall be 5 inches (127 mm) long and have a 1.503 inch (38.18 mm) maximum diameter.

4/ Focusing electrode voltage shall be measured as follows: With a standard 35-to-100-line raster, modulate grid 1 by 1 microsecond pulses to produce a polka dot pattern. Adjust the beam current to peak value of 100 microamperes. Adjust the focus voltage to obtain the best circular spot at the center of the tube face. A comparator fitted with a reticle having circles approximately the size of the spot facilitates focusing. The focusing voltage obtained in this manner is the focusing electrode voltage.

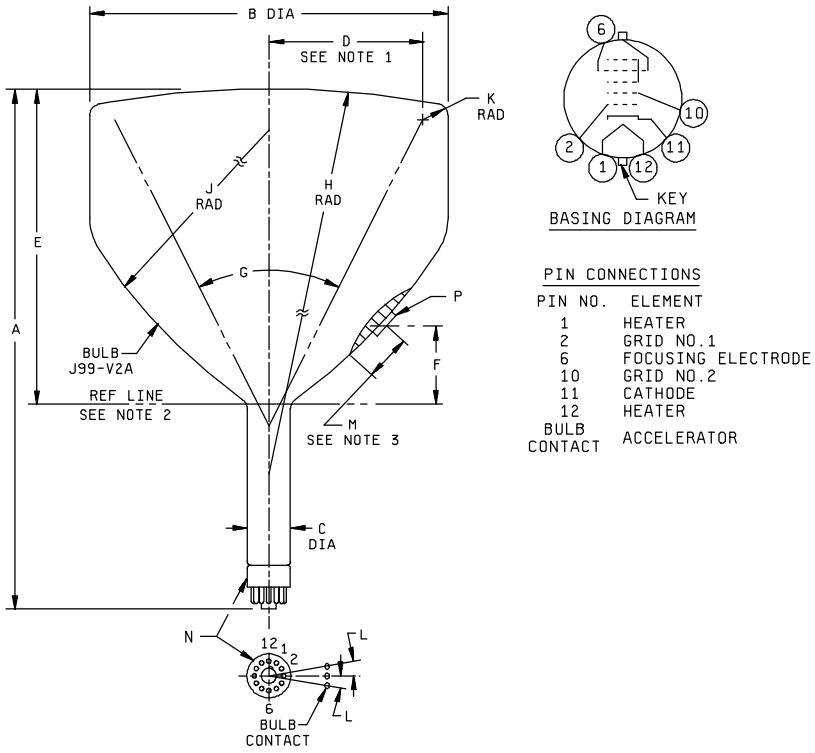
5/ Use an overscanned 35-to-105-line raster pattern with not less than .250 inch (6.35 mm) spacing between adjacent lines; then adjust for best focus at center of tube.

6/ Line width A shall be obtained at the tube orientation producing poorest line width at the center of the tube. The tube position which produces the poorest line width can be found by making the raster nonlinear. This can generally be accomplished by adjustment of vertical linearity control. Measurement of line width A shall be made as follows: Compress the raster until part is merged and part is expanded. Rotate the tube slowly through 180 degrees and note spacing of raster lines, for example: If portions of expanded raster become merged, line width has become poorer; if merged portion of raster becomes expanded, line width has improved.

7/ This test to be performed at the conclusion of the holding period.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:20Z
Check the source to verify that this is the current version before use.




| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| Qualification | | | | |
| L | 10° | | 10° | |
| N | Base: B6-63 | | | |
| P | Bulb contact: J1-21 | | | |
| Conformance inspection, part 2 | | | | |
| A | 17.630 | 18.380 | 447.80 | 466.85 |
| B | 12.310 | 12.560 | 312.67 | 319.02 |
| C | 1.375 | 1.500 | 34.93 | 38.10 |
| D | 5.500 | --- | 139.70 | --- |
| E | 10.310 | 10.690 | 261.87 | 271.53 |
| F | 2.630 | 3.380 | 66.80 | 85.85 |
| M | 1.500 | --- | 38.10 | --- |
| Reference dimensions (see note 4) | | | | |
| G | 55° | | 55° | |
| H | 40 | | 1016 | |
| J | 20 | | 508 | |
| K | .438 | | 11.13 | |

NOTES:
1. Useful screen radius.
2. Reference line is determined by point where gauge G-112 will rest on bulb cone.
3. Anicorna coating. Tube should not be handled by this part of the bulb.
4. Reference dimensions are for information only and are not required for inspection purposes.
5. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube types 12ABP1A, 12ABP7A, and 12ABP19A.



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:20Z
Check the source to verify that this is the current version before use.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Air Force - 99

Preparing activity:
DLA - CC

(Project 5960-3552)



Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:20Z
Check the source to verify that this is the current version before use.